

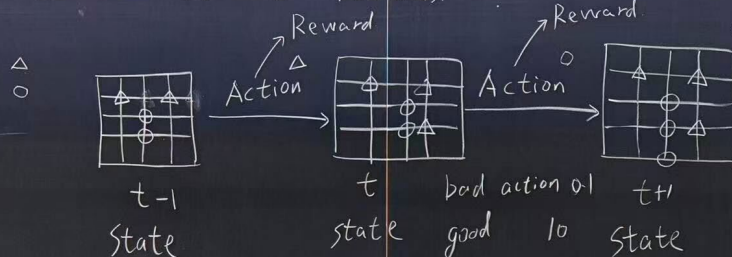
Large language models (LLMs)

- ① Model design $\begin{cases} \text{Language model} \\ \text{Transformer} \end{cases}$
- ② Training : maximum likelihood.
- ③ Use $\begin{cases} \text{Sampling} \\ \text{Prompting} \end{cases} \begin{cases} \text{Zero-shot} \\ \text{Few-shot} \\ \text{chain-of-thoughts} \end{cases}$

- ④ Finetuning $\begin{cases} \text{Supervised finetuning} \\ \text{Reinforcement learning from human feedback (RLHF)} \end{cases}$

$$\max_{\theta} \log \prod_{n=1}^N P(y_n | x_n)$$

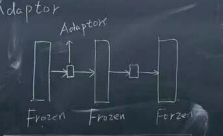
States Actions Rewards





- Low-rank adapt

$\phi(\theta)$
 { Computational cost
 Overfitting
 Parameter-efficient finetuning
 { Adaptor
 Low-rank adaptation (LORA)

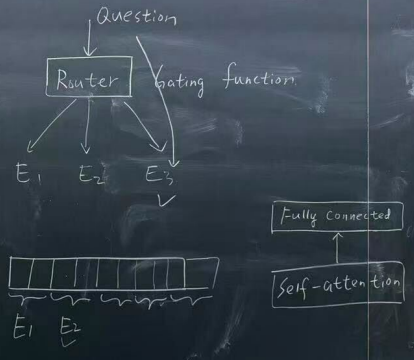
Adaptor

 Frozen Frozen Frozen

PQ
 $d \times r + r \times k$
 $\approx r(d+k)$ $r \ll \min(d, k)$
 $4 \times 20,000 \times 4$
 $A_{d \times k}$
 $10,000 \times 10,000$
 $\downarrow$ frozen trainable
 $\downarrow$ frozen trainable

$d \times k$
 $L + A$
 Pretrained Frozen Update matrix
 $A = P \cdot Q^T (r)$
 $P: d \times r \quad Q: r \times k$

$\max_{\theta} \text{Reward}(\text{action})$
 $\max_{\theta} E_{P(\text{state})} [\text{Reward}(\frac{h_{\theta}(\text{state})}{\theta^*})]$
 $\max_{\theta} P_{\theta}(s_1, a_1, s_2, a_2, \dots, s_n, a_n) \text{ Trajectory}$

Mixture-of-experts (MoE) LLM
 A single powerful LLM
 MoE
 Expert
 LLM LLM
 Math physics Biology


 Question
 Router
 Gating function
 E_1, E_2, E_3
 Fully Connected
 Self-attention

